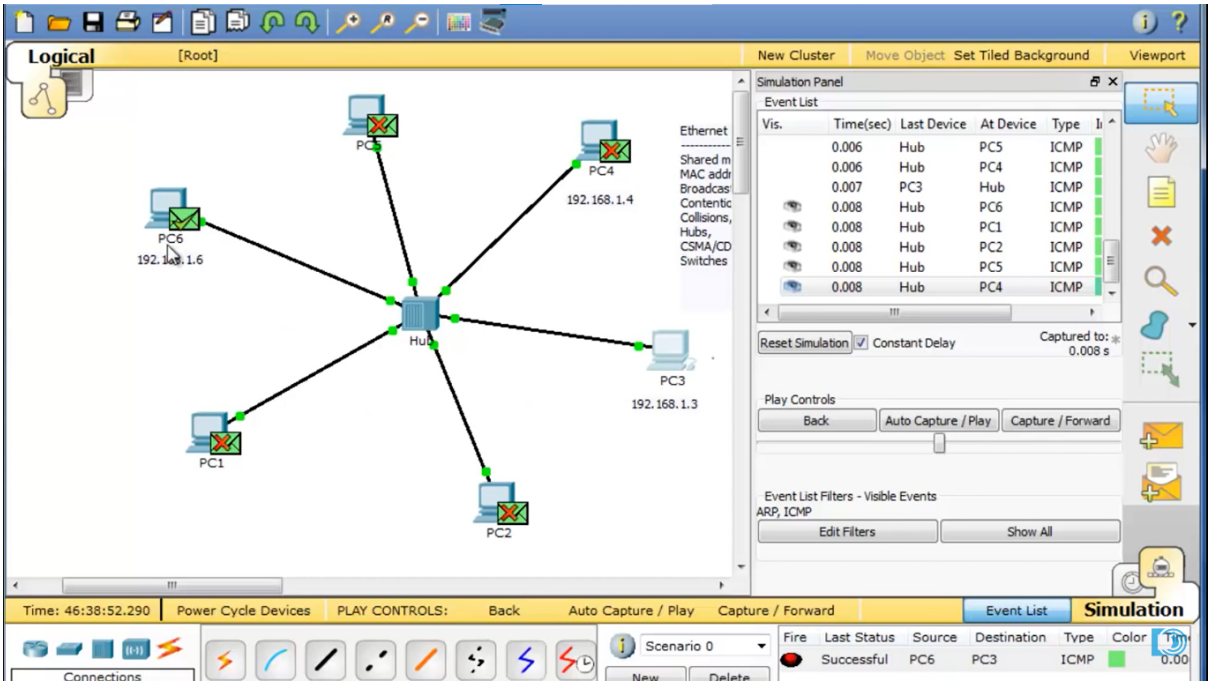
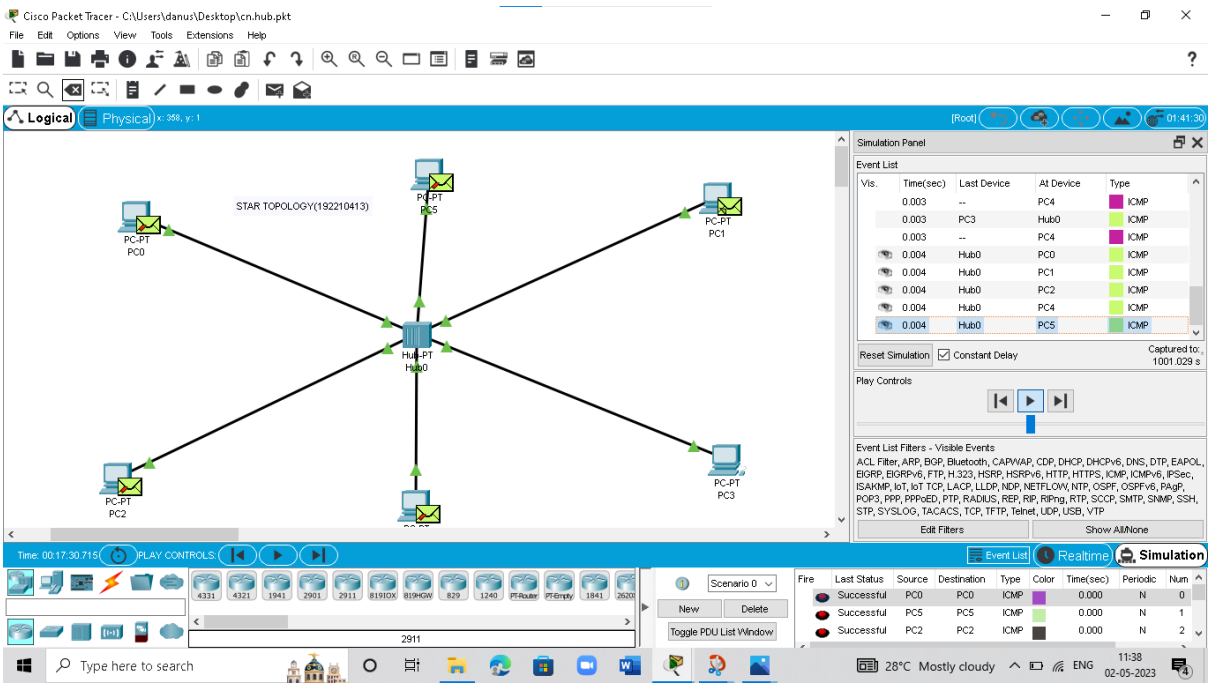
EXPNO:1 Configuration of Network Devices using Packet Tracer tools (Hub, Switch, Ethernet, Broadcast)

Output:



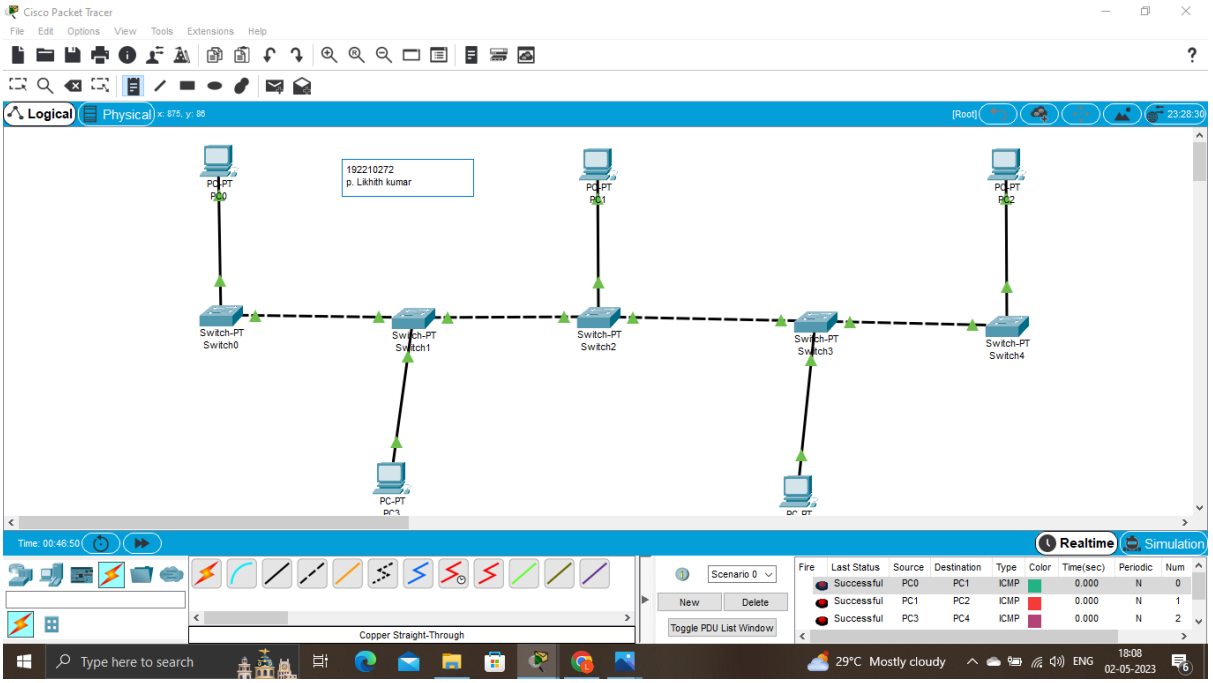
EXPNO:2 Design and Configuration of Star Topologies using Packet Tracer.

Output:



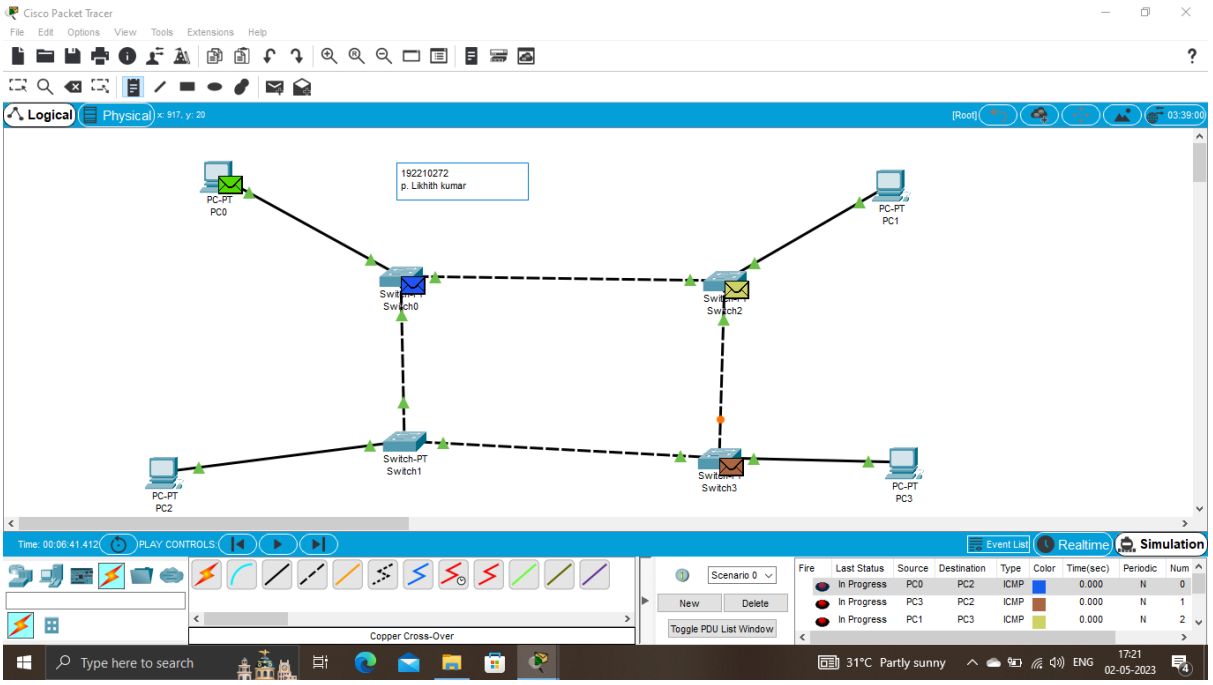
EXPNO: 3 Design and Configuration of BUS Topologies using Packet Tracer.

Output:



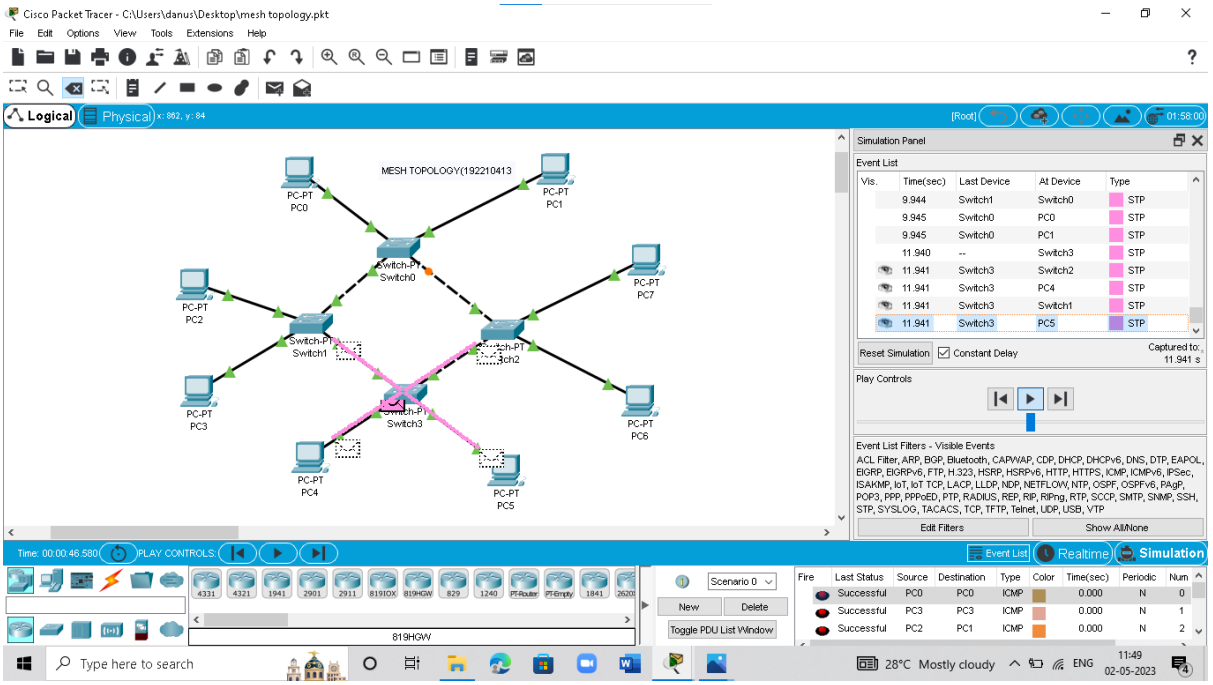
EXPNO:4 Design and Configuration of RING Topologies using Packet Tracer

Output:



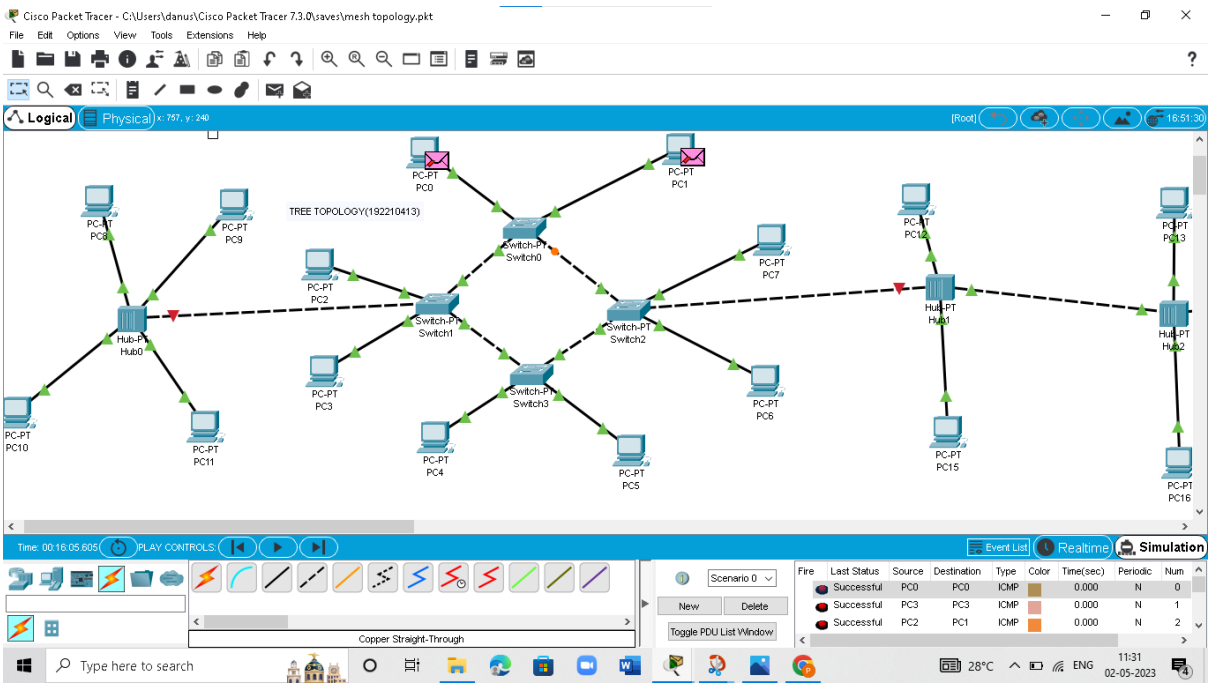
EXPNO:5 Design and Configuration of Mesh Topologies using Packet Tracer

Output:



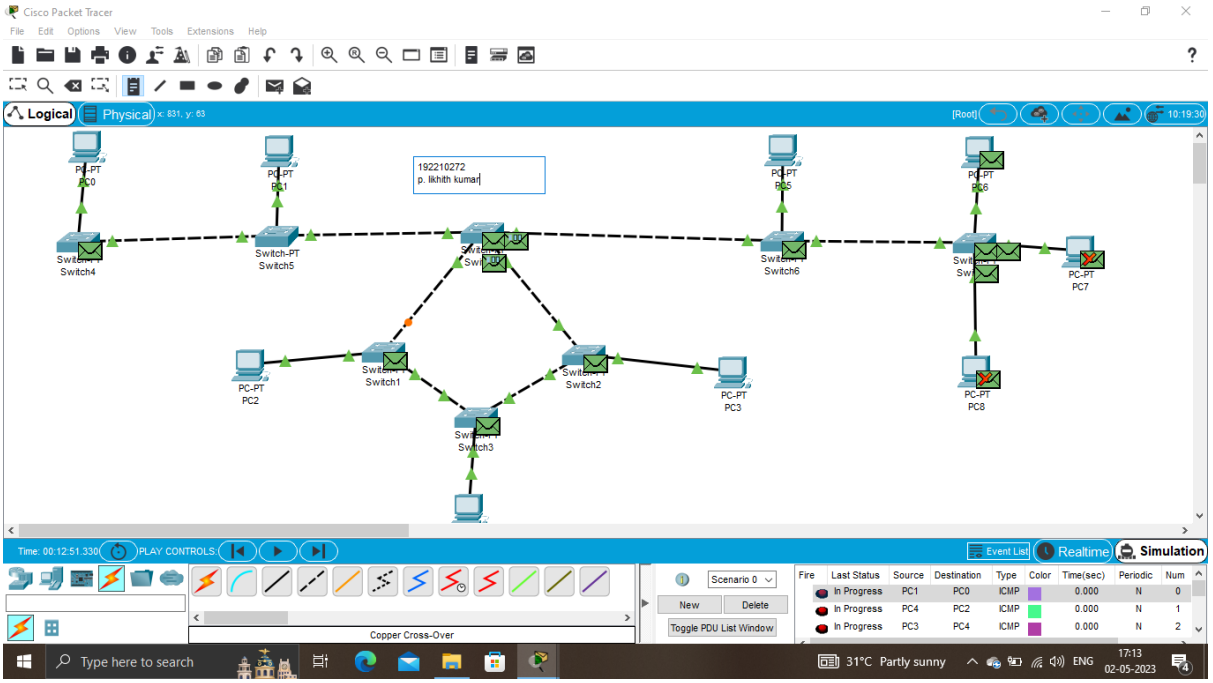
EXPNO:6 Design and Configuration of Tree Topologies using Packet Tracer.

Output:



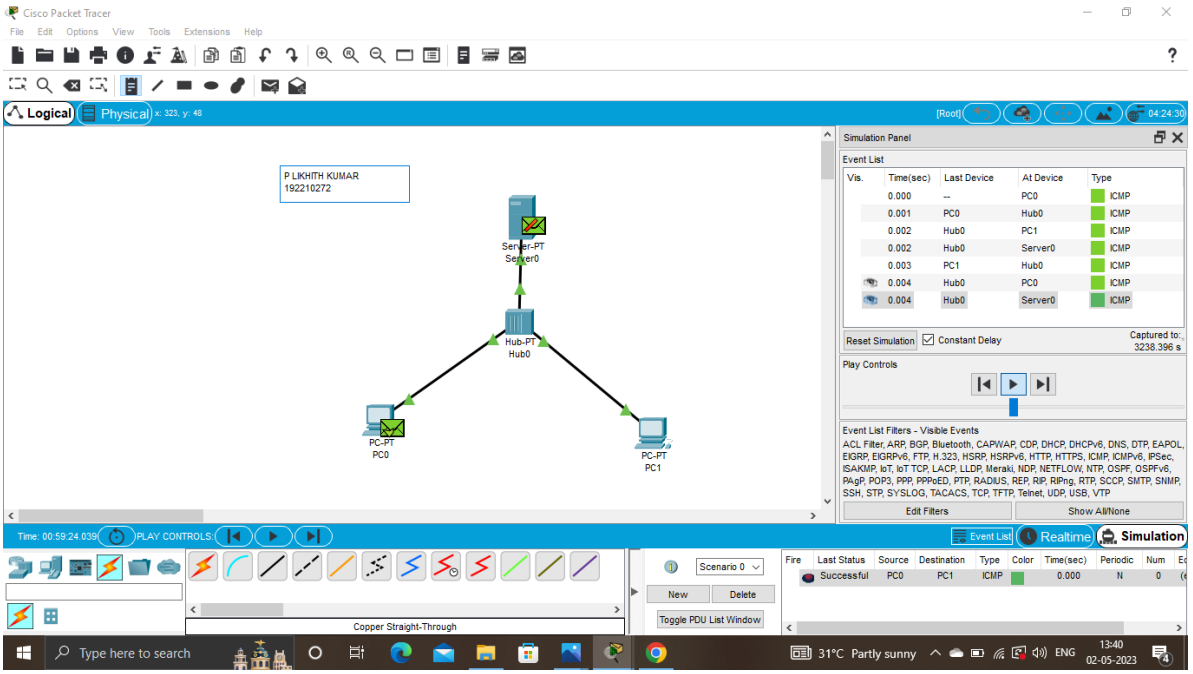
EXPNO:7 Design and Configuration of Hybrid Topologies using Packet Tracer.

Output:



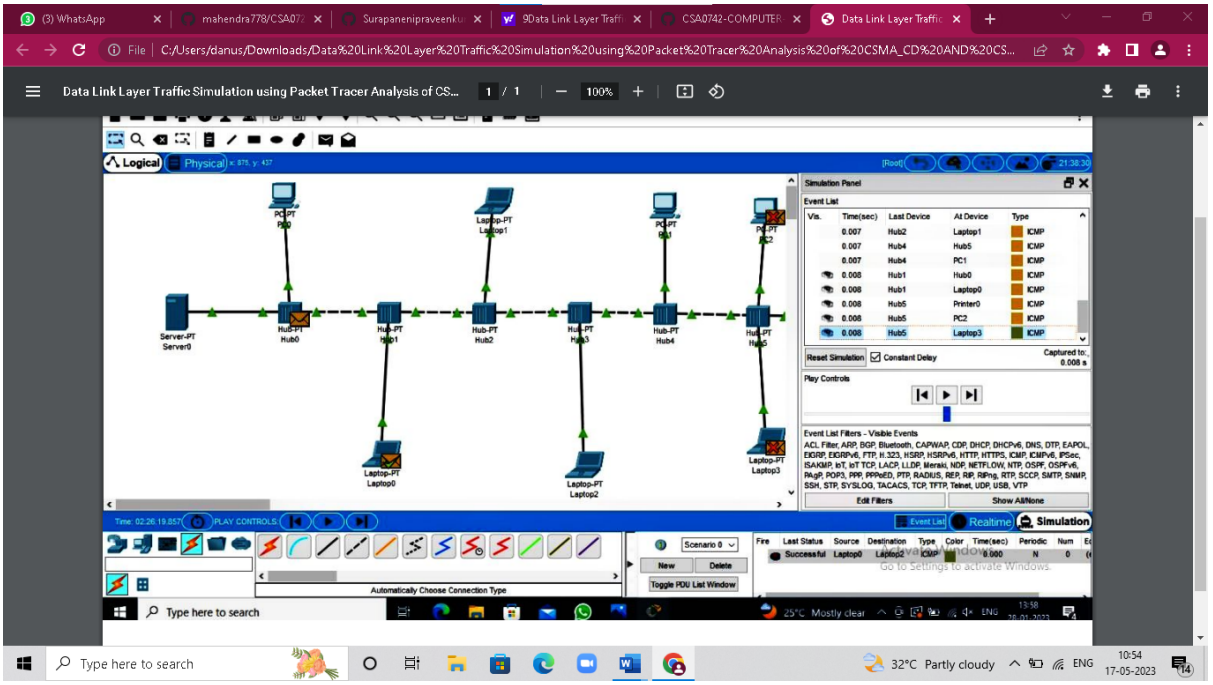
EXPNO:8 Data Link Layer Traffic Simulation using Packet Tracer Analysis of ARP.

Output:



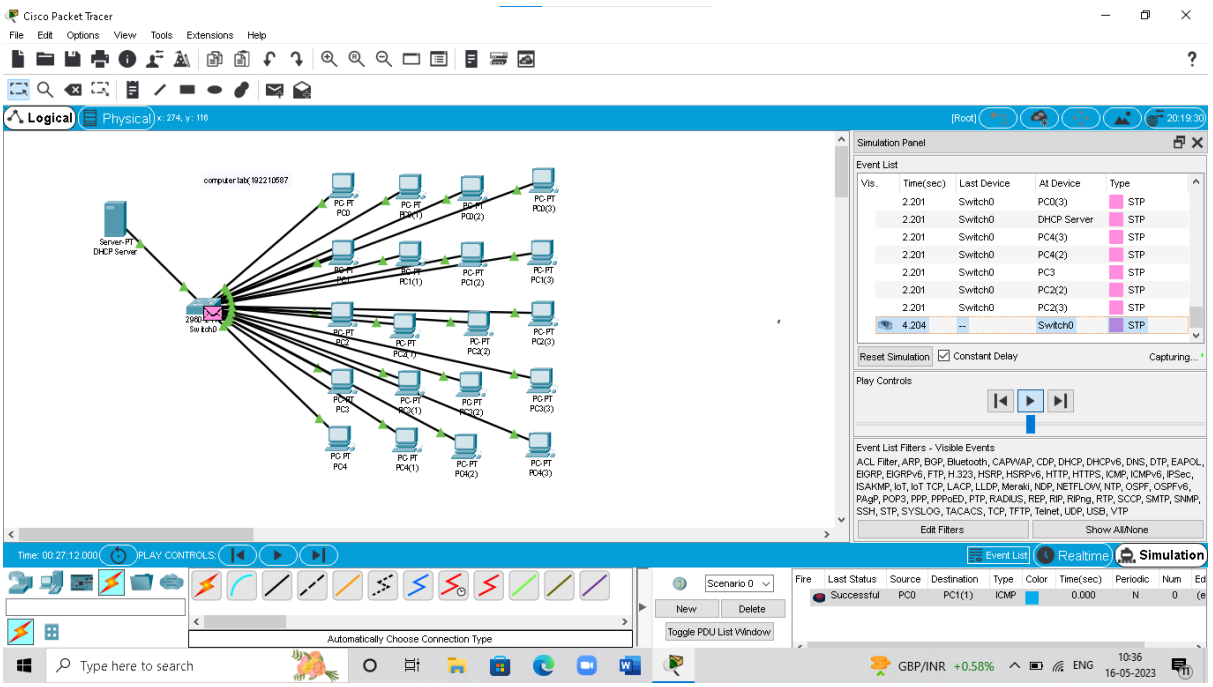
EXPNO:9 Data Link Layer Traffic Simulation using Packet Tracer Analysis of CSMA/CD & CSMA/CA.

Output:



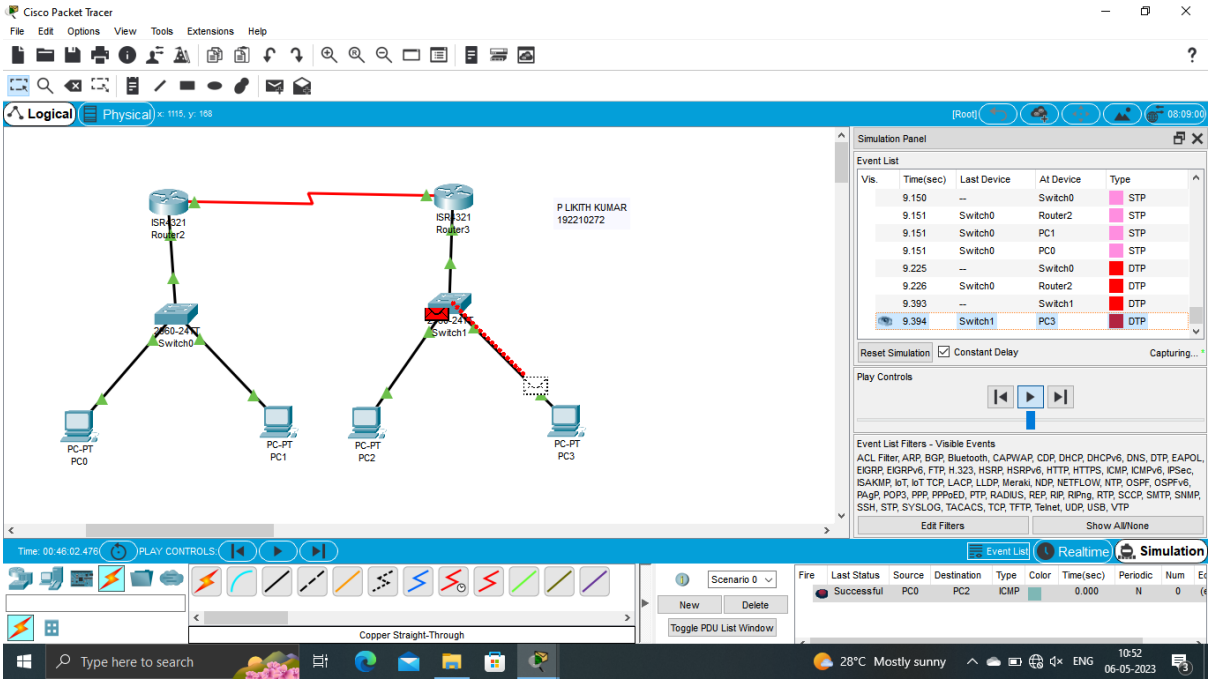
EXPNO:10 Making Computer Lab in Cisco Packet Tracer

Output:



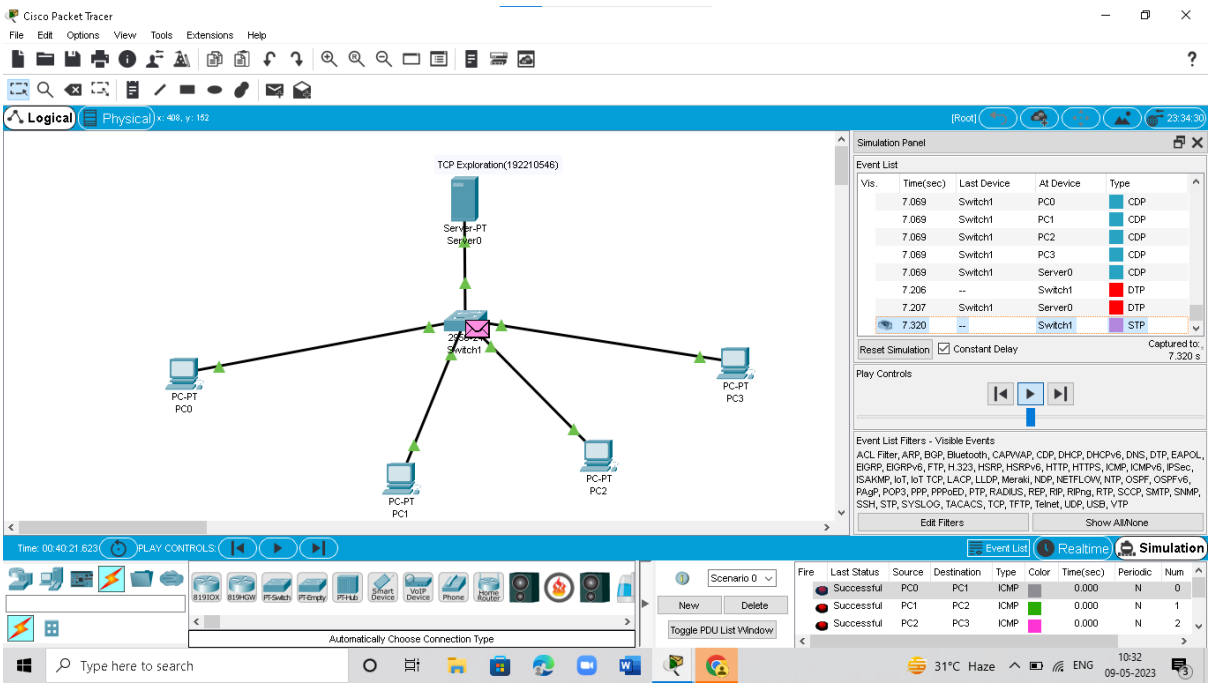
EXPNO:11 Designing two different network with Static Routing techniques using Packet Tracer.

Output:



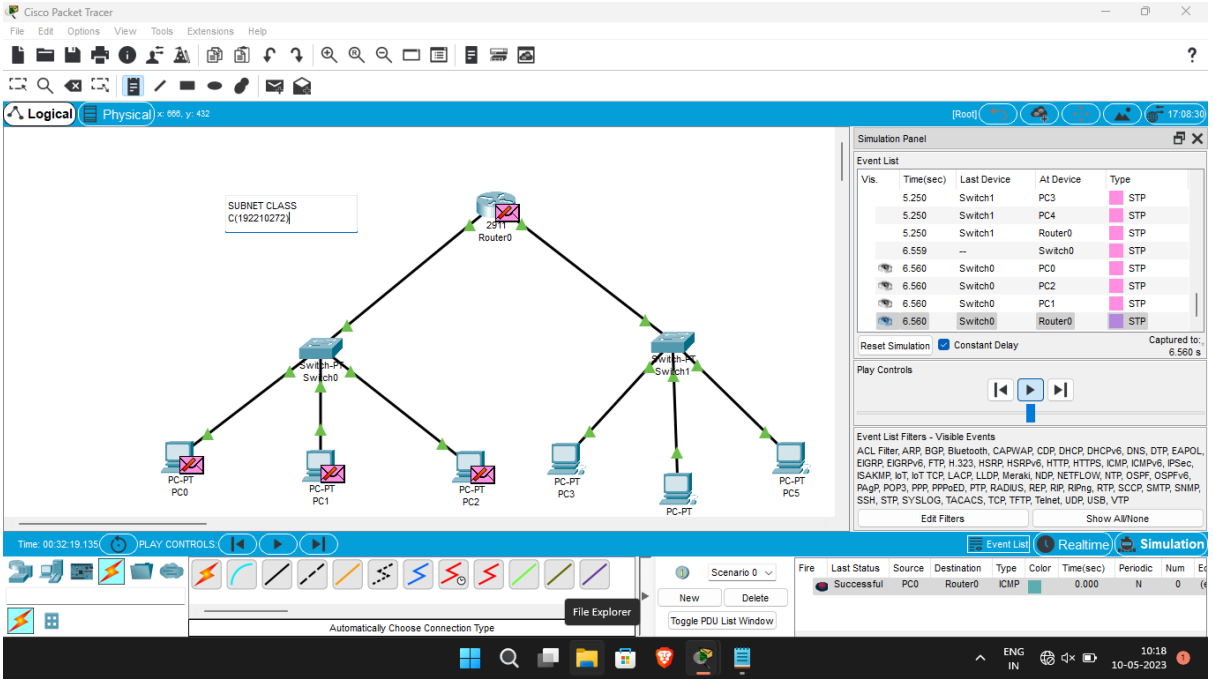
EXPNO:12 Design the Functionalities and Exploration of TCP using Packet Tracer

Output:



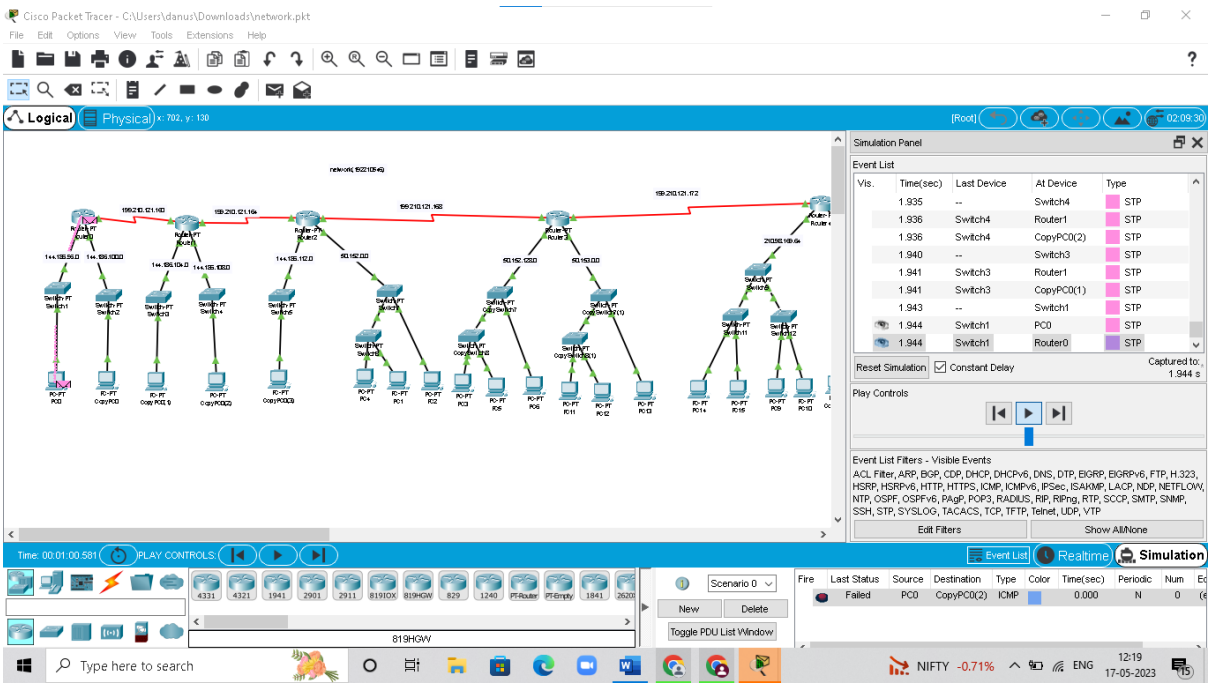
EXPNO:13 Design the network model for Subnetting – Class C Addressing using Packet Tracer.

Output:



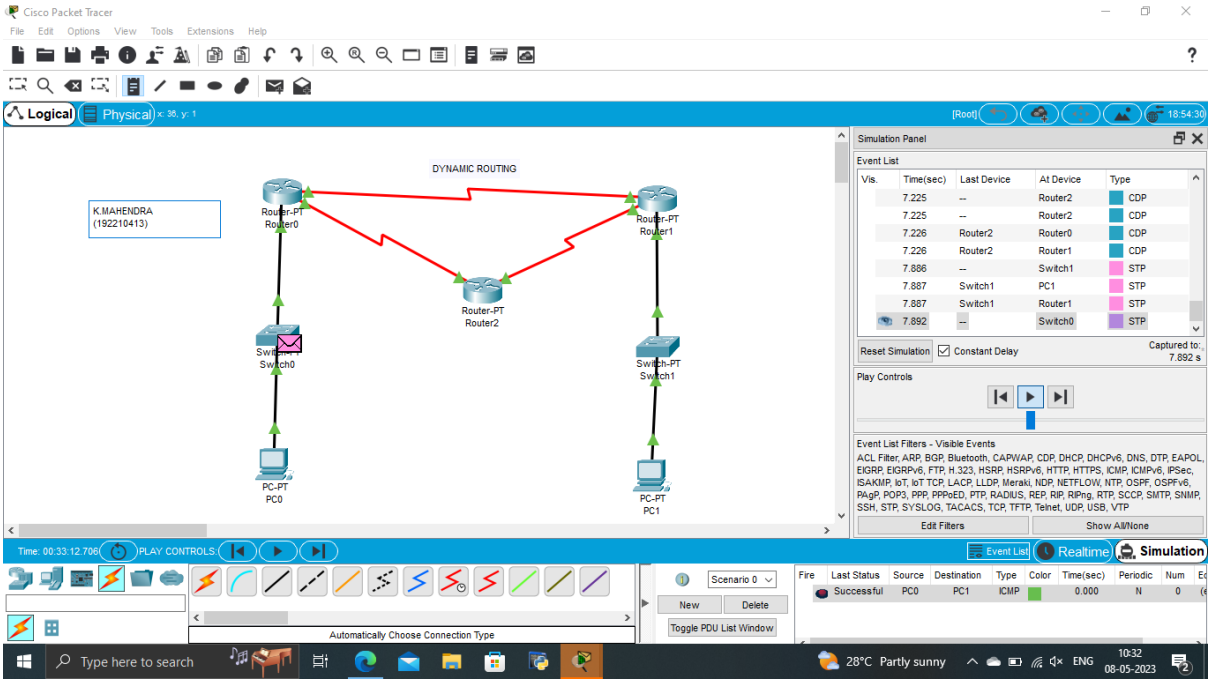
EXPNO:14 Simulating X, Y, Z Company Network Design and simulate using Packet Tracer.

Output:



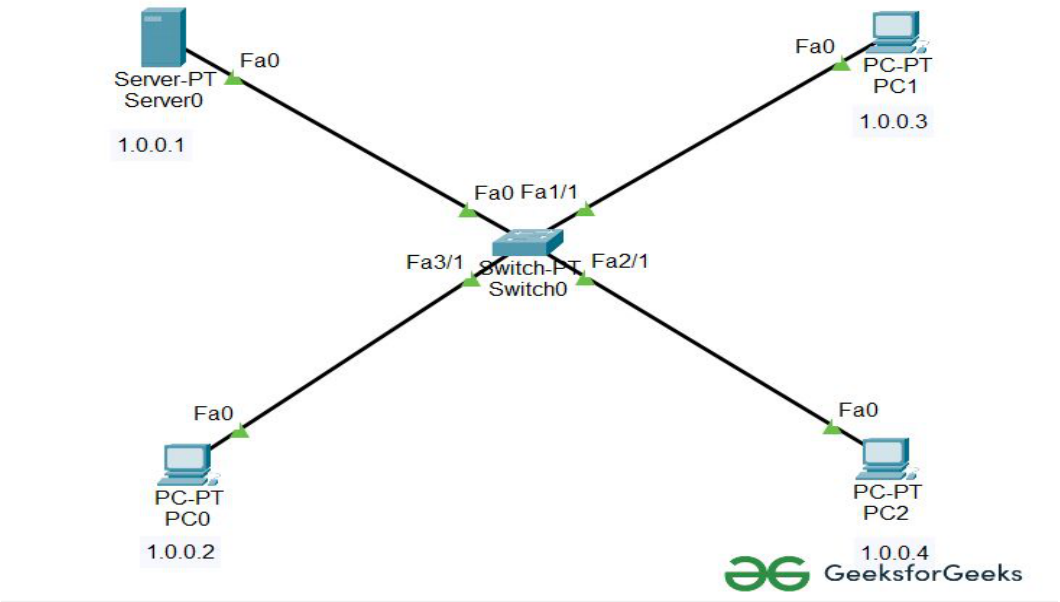
EXPNO:15 Configuration of DHCP (dynamic host configuration protocol) in packet Tracer.

Output:



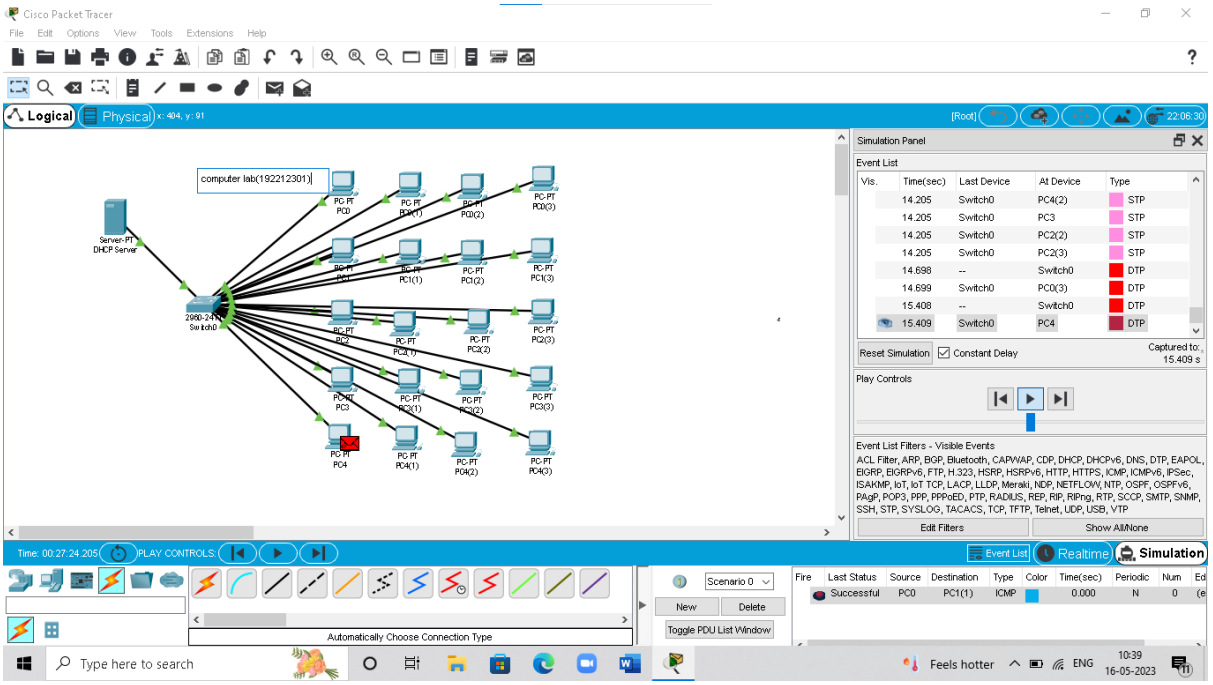
EXPNO:16 Configuration of firewall in packet tracer.

Output:



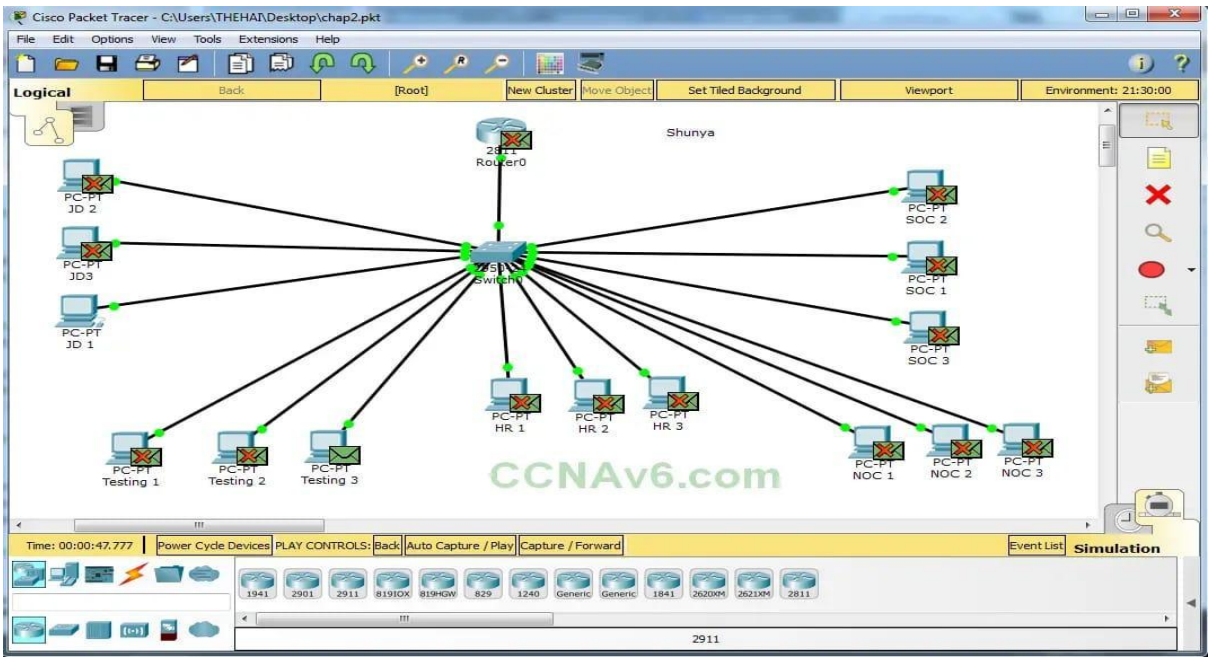
EXPNO:17 Make a Computer Lab to transfer a message from one node to another to design and simulate using Cisco Packet Tracer.

Output:



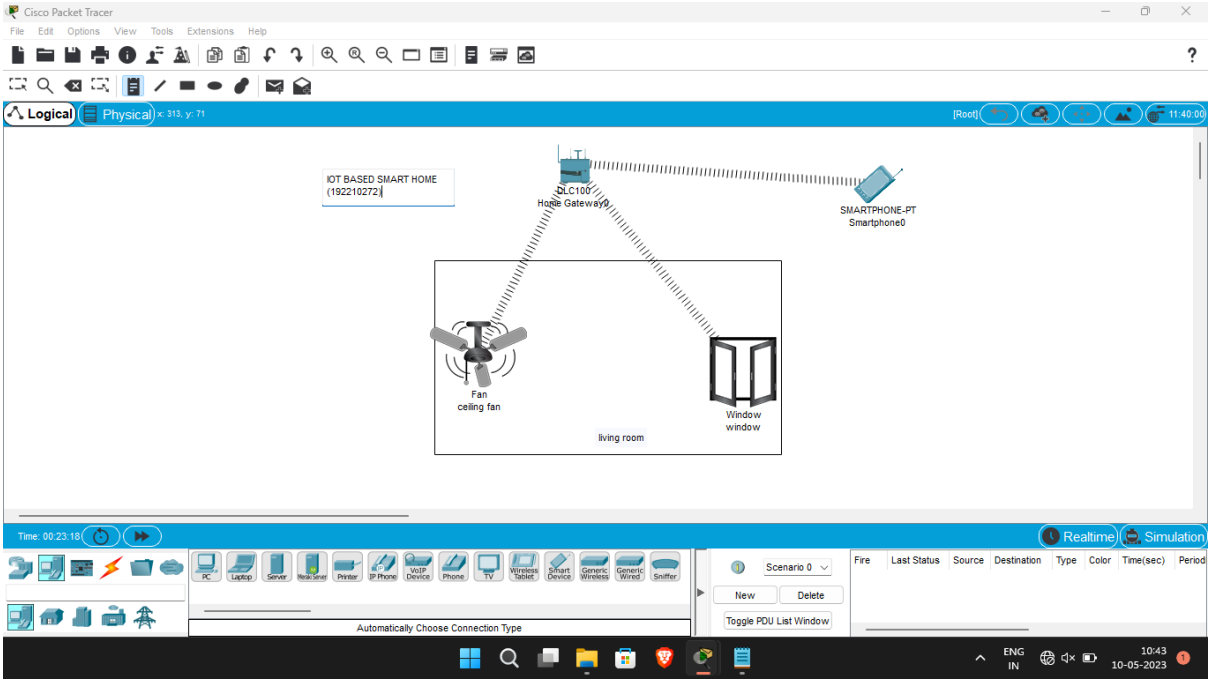
EXPNO:18 Simulate a Multimedia Network in Cisco Packet Tracer.

Output:



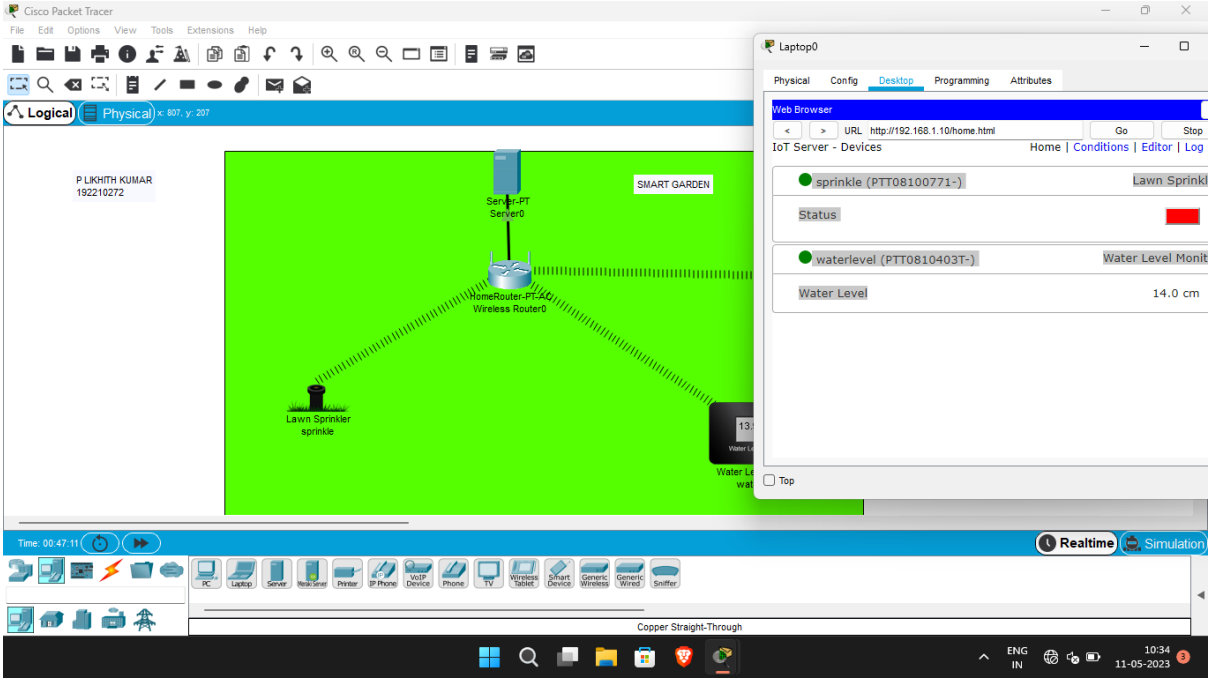
EXPNO:19 IoT based smart home applications.

Output:



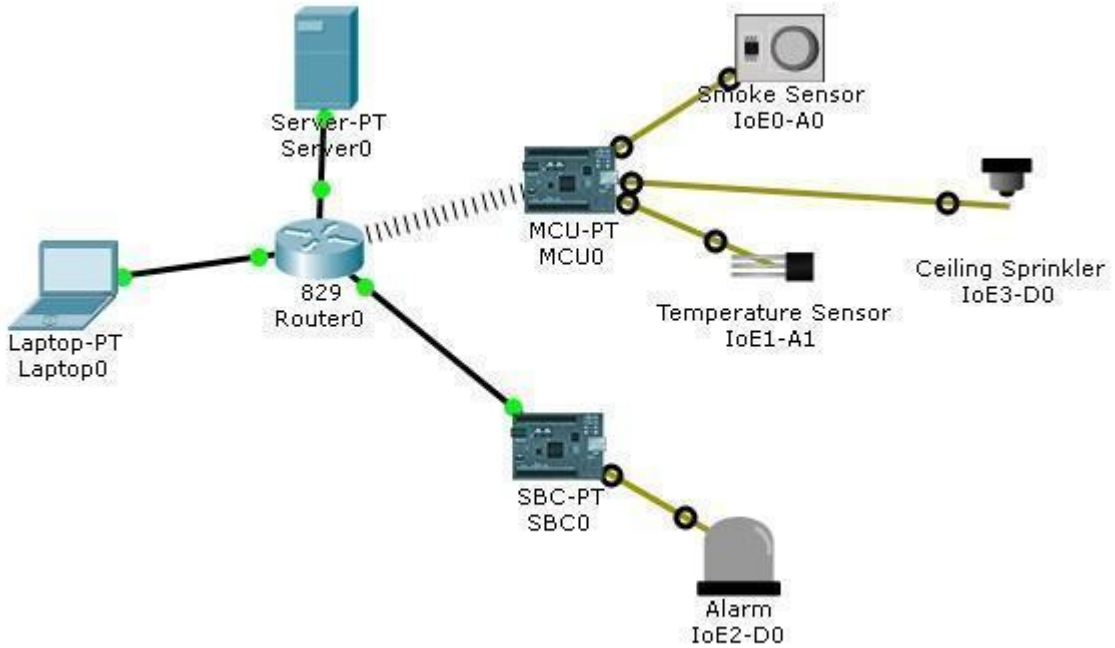
EXPNO:20 Implementation of IoT based smart gardening.

Output:



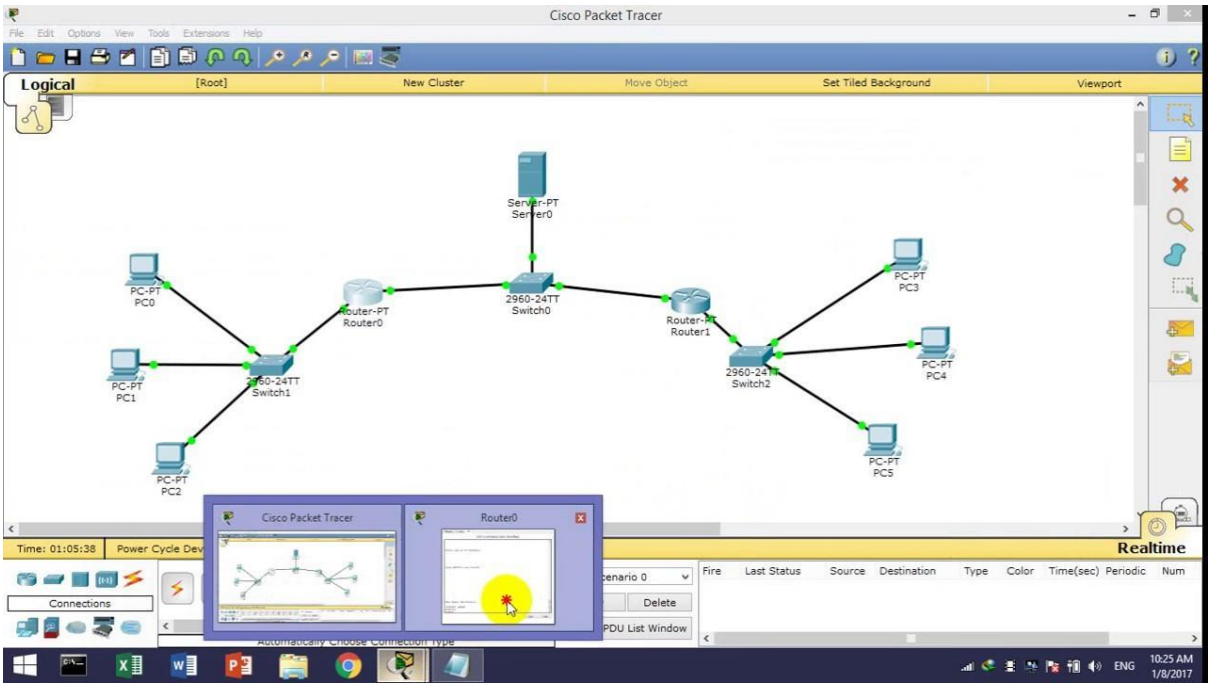
EXPNO:21 Implementation of IoT devices in networking.

Output:



EXPNO:22 IoT based AAA Local and Server based authentication configuration.

Output:


Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Server-PT Server0
PC-PT PC0
Router-PT Router0
2960-24TT Switch0
Router-PT Router1
PC-PT PC3
PC-PT PC1
2960-24TT Switch1
2960-24TT Switch2
PC-PT PC4
PC-PT PC2
PC-PT PC5
Cisco Packet Tracer
Router0
Time: 01:05:38
Power Cycle Dev
Realtime
Connections
Delete
PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
ENG
10:25 AM
1/8/2017

```c
    printf("Enter number of bits: ");
    scanf("%d", &n);

    printf("Enter the bits (0/1):\n");
    for (i = 0; i < n; i++) {
        scanf("%d", &data[i]);
    }

    bitStuffing(data, n);

    printf("Enter stuffed data length: ");
    int stuffed_len;
    scanf("%d", &stuffed_len);
    int stuffed_data[100];
    printf("Enter the stuffed bits (0/1):\n");
    for (i = 0; i < stuffed_len; i++) {
        scanf("%d", &stuffed_data[i]);
    }

    bitDeStuffing(stuffed_data, stuffed_len);

    return 0;
}
```

```
Enter number of bits: 8
Enter the bits (0/1):
0 1 1 1 1 1 1 0
Data after Bit Stuffing: 011111010
Enter stuffed data length: 9
Enter the stuffed bits (0/1):
0 1 1 1 1 1 0 1 0
Data after Bit De-stuffing: 01111110

=== Code Execution Successful ===
```

Time	Source	Destination	Protocol	Length	Info
136 11.867216600	192.168.1.6	52.178.17.234	TLSv1.3	321	Application Data
159 12.061576100	52.178.17.234	192.168.1.6	TCP	58	443 → 55899 [ACK] Seq=6914 Ack=4635 Win=4194816 Len=0
160 12.061576100	52.178.17.234	192.168.1.6	TLSv1.3	105	Application Data
161 12.061576100	52.178.17.234	192.168.1.6	TLSv1.3	502	Application Data
162 12.061814300	192.168.1.6	52.178.17.234	TCP	54	55899 → 443 [ACK] Seq=4635 Ack=6961 Win=261376 Len=0
163 12.061903300	192.168.1.6	52.178.17.234	TCP	54	55899 → 443 [ACK] Seq=4635 Ack=7405 Win=260864 Len=0
227 17.378978500	192.168.1.6	148.113.17.97	TCP	55	[TCP Keep-Alive] 58299 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
228 17.488380400	148.113.17.97	192.168.1.6	TCP	70	[TCP Keep-Alive ACK] 443 → 58299 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
613 23.913297600	192.168.1.6	13.204.106.34	TCP	66	57161 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
615 23.971965900	13.204.106.34	192.168.1.6	TCP	70	443 → 57161 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1250 SACK_PERM WS=4096
616 23.972199300	192.168.1.6	13.204.106.34	TCP	54	57161 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
617 23.973114700	192.168.1.6	13.204.106.34	TCP	1304	57161 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1250 [TCP PDU reassembled in 618]
618 23.973114700	192.168.1.6	13.204.106.34	TLSv1.3	567	Client Hello (SNI=threat.api.mcafee.com)
619 24.049903300	13.204.106.34	192.168.1.6	TCP	58	443 → 57161 [ACK] Seq=1 Ack=1764 Win=32768 Len=0
620 24.049903300	13.204.106.34	192.168.1.6	TLSv1.3	785	Server Hello,

```
    0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x010 (ACK)
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0x70e6 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (1250 bytes)
    [Reassembled PDU in frame: 618]
    TCP segment data (1250 bytes)
```

```
0030  00 ff 70 e6 00 00 16 03  01 06 de 01 00 06 da 03   ··p····· ........
0040  03 53 2a 1b ea f5 06 49  a3 f5 c0 ad 69 cd f5 84   ·S*····I ····i···
0050  b7 bc a8 04 d8 ec 41 75  ff 0b 73 1b 83 f0 96 5f   ......Au ··s·····_
0060  5f 20 26 38 ac 3f f0 b5  67 0d d9 a6 6c 86 11 c9   _ &8·?·· g···l···
0070  e2 37 67 b4 c8 c6 a8 fb  8f 0d 33 8a 12 45 7c 7f   ·7g····· ··3··E|·
0080  c7 6f 00 20 0a 0a 13 01  13 02 13 03 c0 2b c0 2f   ·o· ···· ·····+·/
0090  c0 2c c0 30 cc a9 cc a8  c0 13 c0 14 00 9c 00 9d   ·,·0···· ........
00a0  00 2f 00 35 01 00 06 71  9a 9a 00 00 00 0b 00 02   ·/·5···q ........
00b0  01 00 00 00 00 00 1a 00  18 00 00 15 74 68 72 65   ........ ····thre
00c0  74 2e 61 70 69 2e 6d 63  61 66 65 65 2e 63 6f 6d   t.api.mc afee.com
00d0  00 17 00 00 00 33 04 cf  04 cd 8a 8a 00 01 00 11   ·····3·· ........
00e0  ec 04 c0 bc 6b 8a c2 d6  61 b3 24 00 3e 94 04 37   ····k··· a·$·>··7
00f0  ea ad 4b 8b 45 6b 32 3f  4c fb 57 2c d9 2a 8a f7   ··K·Ek2? L·W,·*··
0100  3d 82 71 76 c2 48 4e 3b  22 03 71 9b b1 04 d0 a8   =·qv·HN; "·q·····
0110  44 bc 58 aa 25 82 93 6c  c3 2d a7 c5 fb 8c 99 d3   D·X·%··l ........
0120  39 a4 4e 2b b2 80 cb 5d  b9 ea 1a 1a 3c 6a 79 f5   9·N+···] ····<jy·
0130  9d 4b 54 7e f2 76 1c bb  5a 0f 24 1b 3a 92 03 96   ·KT~·v·· Z·$·:···
0140  53 a6 24 8a 50 88 c7 92  a4 88 65 8d b5 a9 61 21   S·$·P··· ··e···a!
```

A data segment used in reassembly of an upper layer protocol (ULP) (up to segment data), 1,250 bytes

Packets: 17100 · Displayed: 3683 (30.2%) · Dropped: 0 (0.0%)

Profile: Default

Search

ENG IN

67%

20:46 19-03-2026

No.	Time	Source	Destination	Protocol	Length	Info
540	59.685152900	148.113.17.97	192.168.1.6	TCP	70	[TCP Keep-Alive ACK] 443 → 58299 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
564	61.943120400	2401:4900:88de:7b43…	2001:4860:4802:34::…	TCP	86	62696 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
565	61.943587700	2401:4900:88de:7b43…	2001:4860:4802:34::…	TCP	86	65260 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
575	61.950976600	2401:4900:88de:7b43…	2404:6800:4007:80f:…	TCP	86	63260 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
576	61.951532000	2401:4900:88de:7b43…	2404:6800:4007:81b:…	TCP	86	50036 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
579	61.957968200	2401:4900:88de:7b43…	2404:6800:4000:1025…	TCP	86	54844 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
580	61.958491100	2401:4900:88de:7b43…	2404:6800:4000:1025…	TCP	86	55331 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
581	61.967255200	2001:4860:4802:34::…	2401:4900:88de:7b43…	TCP	90	443 → 62696 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1230 SACK_PERM WS=256
582	61.967681400	2401:4900:88de:7b43…	2001:4860:4802:34::…	TCP	74	62696 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
583	61.967785500	2001:4860:4802:34::…	2401:4900:88de:7b43…	TCP	90	443 → 65260 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1230 SACK_PERM WS=256
584	61.967888100	2401:4900:88de:7b43…	2001:4860:4802:34::…	TCP	74	65260 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
585	61.968732300	2401:4900:88de:7b43…	2001:4860:4802:34::…	TCP	1304	62696 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1230 [TCP PDU reassembled in 586]
586	61.968732300	2401:4900:88de:7b43…	2001:4860:4802:34::…	TLSv1.3	620	Client Hello (SNI=oauthaccountmanager.googleapis.com)
587	61.969656900	2401:4900:88de:7b43…	2001:4860:4802:34::…	TCP	1304	65260 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1230 [TCP PDU reassembled in 588]
588	61.969656900	2401:4900:88de:7b43…	2001:4860:4802:34::…	TLSv1.3	652	Client Hello (SNI=oauthaccountmanager.googleapis.com)

```
    Acknowledgment number (raw): 3366896128
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x018 (PSH, ACK)
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0x9edd [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▸ [Timestamps]
  ▸ [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (546 bytes)
    TCP segment data (546 bytes)
▸ [2 Reassembled TCP Segments (1776 bytes): #585(1230), #586(546)]
```

```
0000  c1 f8 50 18 00 ff 9e dd  00 00 7d 35 00 14 2c 87   ··P····· ··}5··,·
0010  c7 54 2c 5a 55 a4 87 44  10 15 78 41 23 50 95 48   ··,ZU··D ··xA#P·H
0020  a9 07 86 06 4c 29 89 28  56 56 62 50 b2 e1 6b 70   ····L)·( VVbP··kp
0030  8a b0 36 5a cd 4f 89 4d  5a b0 66 ac d5 66 d5 d6   ··6Z·O·M Z·f··f··
0040  5e d0 9a 32 70 67 58 74  32 cd 61 a3 a0 82 a0 bc   ···2pgXt ··a·····
0050  2a b6 67 0f 83 08 a6 27  6a 59 39 80 89 fc 7a 59   ·*·g····' jY9···zY
0060  49 22 d5 d8 af 5e a8 a4  58 8a 9d a9 d4 98 cd 81   I"····^·· X·····
0070  41 c1 74 27 9d ba 18 69  42 87 a8 7c a5 80 a4 4c   A·t'···i B··|···L
0080  a4 f8 c7 d3 10 77 c0 ca  8f f7 67 2b ff b5 8a ea   ·····w·· ··g+····
0090  e7 65 e0 7a 4c b9 e4 63  26 15 b9 2b b0 18 b6 a9   ·e·zL··c &··+····
00a0  a4 0c db 4b 44 66 cb f0  d1 39 8c 40 0f be b8 81   ···KDf·· ·9·@····
00b0  82 dc 64 8f 17 3c 13 67  27 87 bb 0d f7 60 1d 7e   ··d··<·g '····`·~
00c0  e8 9c a9 17 95 60 c6 80  77 04 be 12 52 50 97 db   ·····`·· w···RP··
00d0  37 4f c6 ca c6 db 7e 40  2b 1d 90 30 28 b5 b4 0d   7O····~@ +··0(···
00e0  50 55 c6 bd fb 1f a7 b3  6f 35 b3 49 f5 5c 80 48   PU------ o5·I·\·H
00f0  28 ac 3b fb b8 8e e6 8e  23 c5 a3 cd 24 5e 3d 8e   (·;····· #···$^=·
```

Packet (620 bytes) | Reassembled TCP (1776 bytes)

A data segment used in reassembly of an upper layer protocol (ULP) (fragment: data), 546 bytes

Packets: 1826 · Displayed: 897 (49.1%) · Dropped: 0 (0.0%) Profile: Default

Search

ENG IN 56% 20:44 19-03-2026